

Zixuan Xie

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EDUCATION

University of Virginia (UVA) Ph.D. Candidate in Computer Science Sequential Intelligence Lab (SIL), advised by Professor Shangdong Zhang	09/2024–Present Expected 06/2029; GPA: 3.86/4.0
Fudan University (FDU) B.E. in Software Engineering	09/2020–06/2024 GPA: 3.58/4.0
University of Alberta (UA) Exchange Program Major: Computer Science	01/2023–04/2023 GPA: 4.0/4.0

PUBLICATIONS & PREPRINTS

* indicates equal contribution. Research advisors are underlined.

Preprints

4. *Predict Before You Modify: Learning Improvement Dynamics for Self-Evolving Agents.*
Zixuan Xie, Chang Xu, Yuhua Jiang, Li Zhao.
Preprint, 2026.
3. *Convergence and Emergence of In-Context Reinforcement Learning with Chain of Thought.*
Zixuan Xie, Xinyu Liu, Rohan Chandra, Shangdong Zhang.
Preprint, 2026.
2. *Beyond Linear Attention: Softmax Transformers Implement In-Context Reinforcement Learning.*
Zixuan Xie, Xinyu Liu, Claire Chen, Shuze Daniel Liu, Rohan Chandra, Shangdong Zhang.
Preprint, 2026.
1. *Almost Sure Convergence Rates and Concentration of Stochastic Approximation and Reinforcement Learning with Markovian Noise.*
Xiaochi Qian*, **Zixuan Xie***, Xinyu Liu*, Shangdong Zhang.
Preprint, 2024. Under review at Journal of Machine Learning Research (**JMLR**).

Refereed Conference Papers

6. *MathlibPR: Pull Request Merge-Readiness Benchmark for Formal Mathematical Libraries.*
Zixuan Xie*, Xinyu Liu*, Shangdong Zhang.
Neural Information Processing Systems (**NeurIPS**), Evaluations & Datasets Track, 2026. [\[Dataset\]](#)
5. *Almost Sure Convergence Rates of Stochastic Approximation and Reinforcement Learning via a Poisson-Moreau Drift.*
Xinyu Liu, **Zixuan Xie**, Shangdong Zhang.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026.
4. *MathlibLemma: Folklore Lemma Generation and Benchmark for Formal Mathematics.*
Xinyu Liu, **Zixuan Xie**, Amir Moeini, Claire Chen, Shuze Daniel Liu, Yu Meng, Aidong Zhang, Shangdong Zhang.
International Conference on Machine Learning (**ICML**), 2026. [\[Code\]](#)
3. *Offline Two-Player Zero-Sum Markov Games with KL Regularization.*
Claire Chen, Yuheng Zhang, Xinyu Liu, **Zixuan Xie**, Shuze Daniel Liu, Nan Jiang.
International Conference on Machine Learning (**ICML**), 2026.
2. *Finite Sample Analysis of Linear Temporal Difference Learning with Arbitrary Features.*
Zixuan Xie*, Xinyu Liu*, Rohan Chandra, Shangdong Zhang.
Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
1. *Linear Q-Learning Does Not Diverge in L^2 : Convergence Rates to a Bounded Set.*
Xinyu Liu*, **Zixuan Xie***, Shangdong Zhang.
International Conference on Machine Learning (**ICML**), 2025.

Refereed Journals

2. *Efficient Algorithms for Computing Random Walk Centrality.*
Changan Liu, **Zixuan Xie**, Ahad N. Zehmakan, Zhongzhi Zhang.
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2025.
1. *Combinatorial Properties for a Class of Simplicial Complexes Extended from Pseudo-fractal Scale-free Web.*
Zixuan Xie, Yucheng Wang, Wanyue Xu, Liwang Zhu, Wei Li, Zhongzhi Zhang.
Fractals, 2023.

RESEARCH EXPERIENCE

Microsoft Research Asia (MSRA) | Research Intern
Machine Learning Group, mentored by Chang Xu

05/2026–09/2026

Learning Improvement Dynamics for Self-Evolving Agents

2026

- Designed and implemented an editable Improvement Dynamics Model (IDM) that predicts task-level outcomes of proposed agent modifications, guides selective revision, and learns from forecast–outcome records.
- Built evolution and evaluation pipelines across coding, paper review, robotics reward design, and mathematical grading, including held-out evaluation and cross-domain predictor transfer.
- Improved median area under the performance–budget curve on Polyglot by 46% over HyperAgents across 40 generations and by 10.9% under matched token budgets including evolution and task evaluation.

University of Virginia | Ph.D. Research

09/2024–Present

Sequential Intelligence Lab (SIL), advised by Professor Shangdong Zhang

MathlibPR: Evaluating Contributions to Formal Mathematics

2026

- Built a benchmark of 15,895 build-passing snapshots from 12,063 Mathlib4 pull requests to evaluate merge readiness beyond formal correctness.
- Designed and implemented staged evaluations of four LLMs and two coding agents, finding approval bias and weak recognition of revision progress even with richer context and repository access.

In-Context Reinforcement Learning with Transformers

2025–Present

- Showed how linear Transformers with chain-of-thought unrolling implement multi-step batch TD updates; derived finite-sample guarantees under Markovian data and analyzed the emergence of the corresponding parameters.
- Studied how softmax Transformers implement weighted kernel TD updates, with theoretical analysis of depthwise convergence and empirical evaluation of emergence under reinforcement pretraining.

Reinforcement Learning Theory and Stochastic Approximation

2024–2025

- Established finite-sample convergence guarantees for linear TD(λ) with arbitrary features in discounted and average-reward settings, and bounded-set convergence rates for unmodified linear Q -learning.
- Developed almost sure convergence rates and maximal concentration bounds for contractive stochastic approximation under Markovian noise, with applications to reinforcement learning.

SERVICE

Reviewer: ICLR 2026, AISTATS 2026, ICML 2026, NeurIPS 2026.

SCHOLARSHIPS & HONORS

UVA Provost’s Fellowship

2024–2025

UVA Engineering Distinguished Fellowship

2024–2025

Outstanding Undergraduate Graduate, Fudan University

2024

Fudan-OPTV of China Scholarship

2023–2024

Excellent Student, Fudan University (recommended by Art Troupe)

2022–2023

Meritorious Winner, Interdisciplinary Contest in Modeling (ICM)

2022

Second Prize Scholarship, Fudan University

2021–2022

Excellent Student, Fudan University (recommended by Software School)

2020–2022

Third Prize Scholarship, Fudan University

2020–2021

SKILLS & QUALIFICATIONS

Programming: Python, MATLAB, Java, C.

Research: LLM agent pipelines; benchmark construction; experimental evaluation and statistical analysis.

Scientific writing: L^AT_EX. **Languages:** Mandarin (native), English (professional working proficiency).